REMARKS ♥

PARAMETER : θ	ESTIMATOR : $\hat{\Theta}$	ESTIMATE : $\hat{\theta}$
unknown value	random variable	know value
μ	$\bar{X}$	$\bar{x}$
p	$\hat{P} = \bar{X}$	$\hat{p}$
$\mu_x - \mu_y$	$\bar{X} - \bar{Y}$	$\bar{x} - \bar{y}$

The idea of inferential statistics is to infer about a parameter e.g. μ using an estimator e.g. $\bar{X}$

If $\bar{X}$ is used to estimate μ, its BIAS is define:

$$BIAS(\bar{X}) = E[\bar{X}] - \mu$$

If $E[\bar{X}] = \mu$ then $BIAS(\bar{X}) = 0$. In this case we say that $\bar{X}$ is an unbiased estimator of μ.